4(b). QUEUE USING LINKED LIST PROGRAM

```
#include <stdio.h>
#include <stdlib.h>

// Structure to create a node with data and next pointer
struct node {
        int data;
        struct node *next;
};

struct node *front = NULL;
struct node *rear = NULL;

// Enqueue() operation on a queue
void enqueue(int value) {
        struct node *ptr;
        ptr = (struct node *)malloc(sizeof(struct node));
        ptr->data = value;
        ptr->next = NULL;
        if ((front == NULL) && (rear == NULL)) {
                front = rear = ptr;
        } else {
                rear->next = ptr;
                rear = ptr;
        }
        printf("Node is Inserted:%d \n\n",value);
}

// Dequeue() operation on a queue
int dequeue() {
        if (front == NULL) {
                printf("\nUnderflow\n");
                return -1;
        } else {
                struct node *temp = front;
                int temp_data = front->data;
                front = front->next;
                free(temp);
                return temp_data;
        }
}

// Display all elements of queue
void display() {
        struct node *temp;
        if ((front == NULL) && (rear == NULL)) {
```

```
                printf("\nQueue is Empty\n");
        } else {
                printf("The queue is \n");
                temp = front;
                while (temp) {
                        printf("%d--->", temp->data);
                        temp = temp->next;
                }
                printf("NULL\n\n");
        }
}


int main()
{
        enqueue(10);
        enqueue(20);
        enqueue(30);
        display();
        dequeue();
        display();
}
```

OUTPUT:

Node is Inserted:10

Node is Inserted:20

Node is Inserted:30

The queue is
10--->20--->30--->NULL

The queue is
20--->30--->NULL